

CHAPTER 1. INTRODUCTION

1.1 Introduction

Introduction to your project and your project background as a whole but in brief.

1.2 Problem statement(s)

- Description of problems that directly influence the motives of the project.
- Applicable for the project of improving or solving a specific case.
- Project that is of new creation, re-creation, or individual initiative.

1.3 Objective

- Can be in point form format (use bullets) together with a brief explanation.
- Describe in sequence the objectives to be achieved. E.g. *To propose. . . , To design. . . , To implement. . . , to Evaluate. . . , etc.*

1.4 Scope

Describe every scope involved in your project and give reason(s) for the involvement. Examples: (e.g. children's education), specific users (e.g. children between 5 and 8), other specific entities, specific platforms (e.g. network, OS) etc.- Can be in point form format (use bullets) together with a brief explanation.

1.5 Project Significance

Describe who/what may benefit from the project and how? You can infer it back to your objectives (if applicable).

1.6 Expected Output

Describe what do you expect from your project, the outcome of your research work or findings and the contribution to the knowledge as a whole.

1.7 Report Organisation

- Give a summary of each chapter presented in this report. For example:

Chapter 1: Introduction

This chapter discuss about

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.

Chapter 6: Project Conclusion

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1.8 Summary

Give a summary of this chapter and next activities to be developed.

CHAPTER 2. LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Introduction

Preview to the literature review and project methodology.

2.2 Facts and findings (based on topic)

2.2.1 Domain

Identify domain related with your project with explanations.

2.2.2 Existing System

- Summarize all the reviewed system in a table.
- Identify domain related with your project with explanations.
- Discuss and state your approach and related or past research, references, case study and other finding that relate to your project title. Examples: readings 1 and 3, experiments you get 3 and 4, case study 5 and 6 etc.
- Tagged the source if you refer the approach from published materials.
- Support the approach with statements (from published materials) to justify your fact findings are sound.
- Identify hardware and software used

2.2.3 Technique

- Summarize all the techniques reviewed in a table.
- State other approaches than what you use that you think also applicable and related and give reason to justify why not.

2.3 Project Methodology

- Describe the selected approach or methodology used in your project with justification (Examples: CRISP-DM, Experimental Paradigm and Methods, and relevant methodology)
- Describe the activities that you may do in every stage and relate with your project. Eg. data collection, data pre-processing, etc.
- Describe the experiment measurements and parameters involved.
- Include statements (from published materials) that support the approach you apply.

2.4 Project Requirements

Describe briefly about the project requirements.

2.4.1 Software Requirement

List software requirements in point form. Examples: MS Visual Basic Professional v.6.x, Apache Tomcat and the related for developing application software, MS Project 2000 for project management etc.

2.4.2 Hardware Requirement

List hardware requirements in point form. Examples: PCs, server, devices and storage.

2.4.3 Other Requirements

State other requirements to be used in your project such as you need special lab for project development, photo copy facility, discussion room and etc.

2.5 Project Schedule and Milestones

- Explain your action plan prior to the end of the project. Apply from what you have learnt from project management.
- List and describe stage by stage of your activities.
- Attach your project timeline or Gantt chart (in 1 page view)

2.6 Summary

Summarize the chapter and explain the next activities to be developed.

CHAPTER 3. REQUIREMENT ANALYSIS

3.1 Introduction

Preview to the analysis phase and how it would be developed.

3.2 Problem Analysis

- Investigate and describe current system scenario/situation. Design the state diagram or data flow diagram or activity diagram from your reference(s), showing how the current system(s) or business(s) runs.
- Use statistics or mathematical formulation and validate the models (if using computational analysis).
- Describe in details the process involved in the methodology to solve the problem.

3.3 Requirement analysis

3.3.1 Data Requirement

- What kind of input and interface should be in the system?
- What kind of data to store in the database?
- It can be illustrated by using Data Model or Data Dictionary.

3.3.2 Functional Requirement

- Specify the functions of the system or algorithm.
- Explain on how it works, records, compute, transforms, and transmits within their frames.
- It can be illustrated by using state diagram, DFD, Context diagram or Use case.

3.3.3 Non-functional Requirement

Specify how well the system perform its intended functions. E.g. Quality requirement, Performance- how many computer resources should it use, how accurate should the result be, how much data should it be able to store?

3.3.4 Others Requirement

Describe each of software hardware and requirements that will be used (justification of usage).

3.4 Summary

Summarize the chapter and explain the next activities to be developed.

CHAPTER 4. DESIGN

Note: The following sub-sections are guidelines, select and include only those related and necessary sections based on project type in your report.

4.1 Introduction

Introductory preview to this chapter. Examples: this chapter defines the results of the analysis of the preliminary design and the result of the detailed design.

4.2 High-Level Design

Describe a high-level view of your system's structure or system's interior. You may refine some of the functions below once you are in PSM II.

4.2.1 System Architecture for expert system/DSS/simulation

- Must have the architecture diagram. Define the architecture view of your system. The architecture view can be present in layer, framework, and tier or patent.
- Include an appropriate diagram to explain artificial intelligence techniques in system development.

OR

System Architecture for robotic and automation

- Define the architecture view of hardware. The architecture view can be presented by layer or framework.
- Describe the static view or dynamic view of the application (use diagram such as state or sequence diagram).

4.2.2 User Interface Design for expert system/DSS/simulation

Refine your user interface design (mock UI) that is defined in Chapter 5 of PSM I.

OR

System Architecture for robotic and automation

Refine on software and hardware defined in Chapter 5 of PSM I.

Navigation Design

Define and refine the navigation flow and types of navigation controls.

Input Design for expert system/DSS/simulation

- Define and refine the screens (e.g. types of inputs such as text, numbers, selection box etc.) used to enter information, as well as any forms on which users write or type information.
- Define and refine validation rule for each of input field.

OR

Input Design for robotic and automation

- Define and refine the control panels (e.g. types of inputs such as text, numbers, and selection box etc.) used to enter command to control robot's assembly system.
- Define and refine validation rule for each of input field.

Output Design

Define and refine the types of outputs including detail reports, summary reports, turnaround documents and graphs. Classify your output in term of periodically or ad-hoc basis. For example, daily, monthly, yearly etc.

4.2.3 Database Design

Conceptual and Logical Database Design

- Introduce briefly to the logical data model (LDM) or entity relation- ship diagram (ERD) and what they have to do with the database design.

- Define, refine and construct the entity relationship (ER) diagrams in details with explanation in text on what basis (business rules) applied for every entity relationship that have been defined in the diagram.
- 4Data dictionary and normalization.

OR

Non-database Design

- Logical and conceptual design.
- Define, refine and provide explanation in text on what basis (business rules) applied for every element in the diagram.

4.3 AI Component Design

- Describe about dataset.
- Describe the proposed AI techniques or pseudo-code or algorithm (e.g. Genetic Algorithm, Artificial Neural Networks, Intelligent Agents, etc).
- Specification and diagrams may be further elaborated. Emphasis should be on the logic of the design and the approach to satisfy the requirements.
- Student should be able to answer on how will the system function.

4.4 Software or Hardware Design (if applicable)

- For those use CRISP-DM: Describe in detail of every stages.
- For those develop hardware: Describe in detail of every robot assembly system. The program specification includes information such as pseudo-code, ladder diagram of robot manipulators.

4.5 Summary

Summarize the chapter and explain the next activities to be developed

CHAPTER 5. RESULTS AND DISCUSSION

5.1 Introduction

- Introductory preview to this chapter.
- Briefly describe the activity involved in the testing phase and what is the testing strategy to be adopted in your project.

5.2 Evaluation of AI Techniques used in the project.

Depending on what are the techniques used :

- For Evolutionary Algorithms project:
 - Quality of solution and scalability of algorithms
- For Machine Learning Algorithms project:
 - Prediction/Classification performance – error rate and robustness
- For the Discrete Simulation project
 - Evaluation of Behavioural testing and accuracy
- For the Geospatial project
 - confusion matrix to compute the overall accuracy, producer accuracy, user accuracy, precision, recall and F1-score.
- For NLP systems project (depending on what technique or application used) Techniques:
 - Machine Learning / Deep Learning – accuracy, precision, recall, F1 score, AUC, or may refer to Machine Learning Algorithms.

- NLP models may be also evaluated using behavioural testing.

Applications:

- Parsers – accuracy, coverage, n-best accuracy, cross-brackets.

- Text Summarization – BLEU / Rouge / METEOR score
- Chatbot – various tests may be used such as the natural flow of dialogue, testing the chatbot’s understanding, testing off-topic handling, etc.
- Development of language model or LLM – Perplexity test, Extrinsic evaluation, Intrinsic evaluation

5.3 Testing of Functional Requirement

- Explain the testing of all the functional requirements of the product.
 - Test case identification, tester identification, test case results (Success/Fail), and detailed documentation on the failed test case.

OR

- For the Robotics project

Testing of Functional Requirements:

- behavioural testing, accuracy;
- confusion matrix, accuracy, precision, recall, F1-score, AUC, or may refer to standard algorithm's metrics;
- classification performances - error rate, robustness;
- others.

Testing of Non-Functional Requirements:

- usability, practicability;
- commercial values;
- design impacts;
- others.

5.4 Summary

Summarize the chapter and explain the next activities to be developed.

CHAPTER 6. CONCLUSION

6.1 Observation on Weaknesses and Strengths

- State the weaknesses and strengths of your project.
- You also may state other's responses regarding project topics.

6.2 Propositions for Improvement

- Present your suggestions on how your system can be improved better.
- Elaborate each of your suggestions in a paragraph.

6.3 Project Contribution

- State your project contribution to the university/faculty/company/individual.
- State where to find the user manual - e.g. Appendix XX

6.4 Summary

- State whether you think your project meets your set objectives conclusively.
- Concluding phrases to conclude the project.

REFERENCES

- A list of references used in the project report such as journals, articles, books and etc. as directly quoted in the report.
- Minimum of 20 references. Please use Harvard styles.
- For complete reference, please refer to Final Year Project Writing Guide 2019.

APPENDICES

- E.g. user guide, user manual, diagrams etc.